Statistical Report

Determinants of Belgian Household Energy Consumption

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1 Background

The goal of "*Carbon Neutrality*" is gaining increasing momentum in the European political arena amid substantial indicators of an incoming climate change which might have deleterious effects at the societal, economic and, most importantly, environmental level. Household energy consumption plays a pivotal role in overall social sustainability since it accounts for a substantial share of the total energy demand in most developed countries. Within this sector, electricity consumption provides a useful proxy for overall household energy demand and can be put into relationship to structural characteristics of the buildings as well as the socio-economic profile of their occupants. This cross-sectional study collected data from 2500 households in Belgium and examines the relationship between households electricity consumption and several predictors.

2 Data management

2.1 Data Cleaning

Three observations (rows 66, 36, and 135) were removed from the dataset according to the instructions. Subsequently, the dataset was cleaned by excluding observations with missing values ,which were 0.2% in six variables including the `annual_kwh` variable.(see Appendix Table 5). After ordering the data in decreasing order, Five observations in the `annual_kwh` variable (rows number: 267, 1330, 1570, 369, 12) displayed too large values (172 to 312 times larger) and, therefore, were removed as were attributed to imputation error (see section 6 for details).

2.2 Data Coding

Variables were coded according to their type: continuous, count, ordinal, categorical, or binary. Table 3 summarizes their descriptions and coding schemes.(Appendix)

3 Descriptive Statistics

3.1 Categorical variables

The distribution of EPC labels was relatively balanced across categories. Labels A (22.2%) and C (21.3%) are the most common, followed by B (17.6%) and D (15.7%). Lower-efficiency categories E and F occur less frequently, 13.6% and 9.6%, respectively. Buildings construction periods and occupancy count were evenly distributed in the sample, with approximately 33% in each building era and 14-15% in each occupancy count. A breakdown is provided in the appendix. None of the categories appear too sparse, suggesting that all variables can be retained for modelling without concerns about low cell counts. From the exploratory analysis, two categorical variables seem to correlate strongly with electricity consumption: EPC label and occupancy count. The other categorical variables did not display an apparent correlation with electricity consumption. Moreover, EPC label and building era are correlated with each other since the most efficient labels are clustered in the two most recent building eras. (see section 6.2).

3.2 Continuous variables

On average, households consume 5,874.76 kWh of electricity annually, with a relatively large standard deviation (2,224.25 kWh). The median consumption (5,432.07 kWh) is slightly lower than the mean, suggesting a right-skewed distribution, which is also confirmed by the QQ plot showing upward deviations above the diagonal line in the upper tail (see appendix). Floor area averages 158.25(36.50) m^2 with a range from 60.26 m^2 to 341.30 m^2. Household income averages €48,842.90 (22,534.97) with a range from €8,487.91 to €206,367.91. As expected for this type of variable the QQ plot shows strong upper-tail deviation, suggesting positive skewness driven by high-income outliers. A breakdown for each variable is provided in section 6.3.

The correlation matrix reveals that annual electricity consumption has a moderate positive correlation with floor area (r = 0.38) and with income (r = 0.29), suggesting that larger and wealthier households tend to consume more energy. Importantly, two predictors exhibit a strong positive correlation with each other, namely floor area and income (r = 0.72). Distance to Brussels shows virtually no correlation with the other variables, suggesting that location relative to Brussels does not strongly influence household energy use, dwelling size, or income in this dataset. Controlling for the income level, the EPC label distribution appears similar across income quantiles (figure 11) and comparable to the overall EPC distribution in the aggregate dataset. Indoor floor area does not associate with the size of the household since median floor area and interquartile ranges remain relatively stable when controlling for permanent residents, suggesting that larger households do not systematically live in larger dwellings in this dataset (figure 10). (see sections 6.5 and 6.6 of the Appendix)

4 Inferential Statistics

4.1 Regression model selectoin

The initial model contained all variables plus interaction terms between income and number of occupants, EPC label and floor area, EPC label and number of occupants, and income and EPC label. Interactions were selected based on literaturen and the protocol research questions. A quadratic term was considered for `occupancy_count` consistently with a possible decreasing marginal effect of additional occupants. We will refer to the initial model as *saturated*.
Predictor retention was determined using a backward elimination stepwise procedure by fitting the full model then removing the least significant predictors that did not meaningfully contribute to the model fit based on the Bayesian Information Criterion (BIC) and p value, which favours more parsimonious models. At each step, the variable whose removal produced the greatest reduction in BIC was eliminated, and the process continued until no further decrease was observed. We called it *first model*.
After fitting the *first model* the diagnostic plots were checked and showed some violations of the standard assumptions. The residuals vs fitted plot indicates the presence of heteroskedasticity, with the variance of the residuals increasing for higher fitted values, and mild deviations from normality in the Q-Q plot. In this regards, the outcome variable `annual_kwh` was log transformed resulting in an improvement of the diagnostic plots, particularly by yielding a more symmetric residual distribution and enhancing homoskedasticity (fig. 1). The final model retained floor area, EPC label, and the number of occupants as predictors of annual electricity consumption. The interaction terms and other covariates were removed. Nevertheless, `building_era` was retained in the model as it

acts as a confounded being related to both `annual_kwh` and `epc_label`as A and B were basically absent before year 2000 (Table 1 and 2).(see section 6.7 for details).

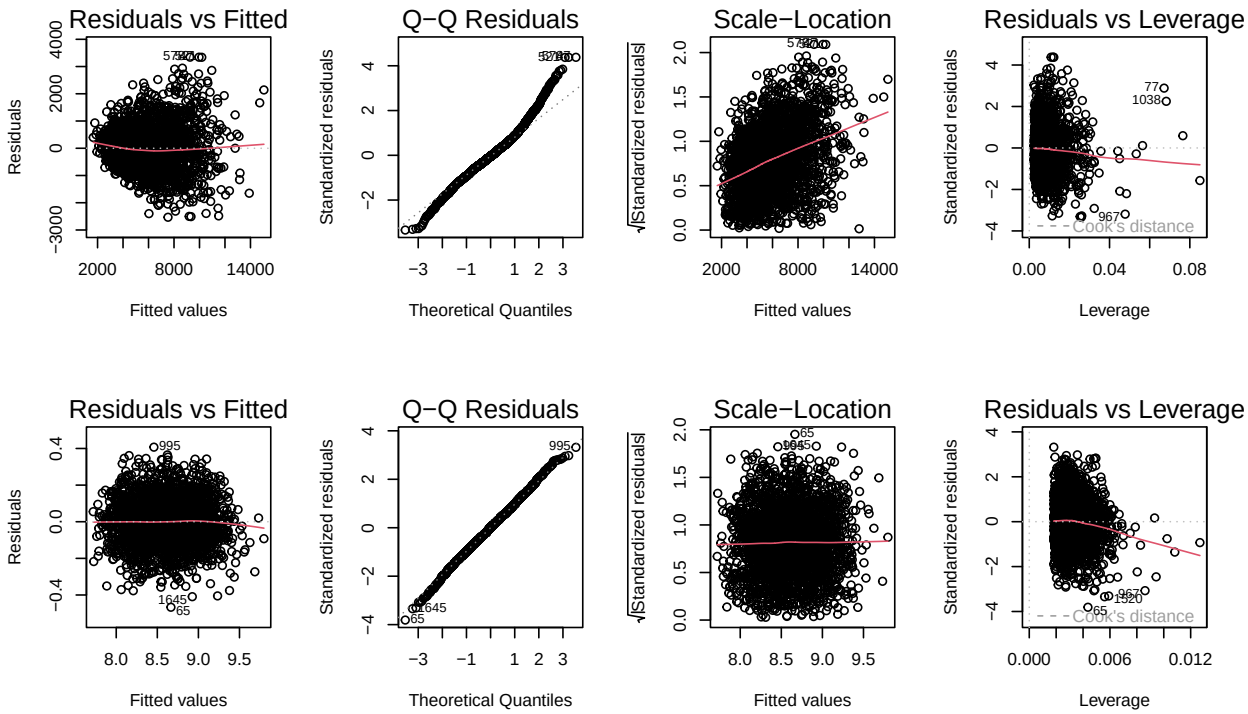


Figure 1: Diagnostic plots for the selected regression model using frist model before (upper plots) and after (lower plots) log-transformation of annual energy consuption.

The final model achieved a substantially lower BIC while maintaining nearly identical adjusted R^2 (89%), indicating that almost all the variance in annual consumption is explained by EPC label, house size and number of occupants. Overall, the diagnostic plots indicate that model assumptions are reasonably satisfied, with only minor deviations that are unlikely to substantially affect inference. the residuals vs leverage plot shows a few observations with higher leverage, but none appear to exert undue influence based on Cook's distance, indicating the absence of strongly influential points. The model was free of multicolinearity and variance inflation Factor (VIF) value were less than 5 (our critical value).

Table 1: Model comparison (initial,log-model, and final) using AIC, BIC, and Adj-R-squared

	Model	AIC	BIC	Adj_R2
initial_model	initial_model	39966.475	40094.400	0.881
model_log	model_log	-3350.746	-3298.412	0.891
final_model	final_model	-3347.589	-3283.627	0.891

4.1.1 Interpretation

The model indicates that, compared to EPC class A, lower efficiency classes exhibit progressively higher electricity consumption. Houses with a 'B' rating use approximately 9 % more energy than houses with the reference label (A), while 'C', 'D', 'E' and 'F' use about 16%, 27%, 37% and 48%, respectively, keeping other predictors fixed. This was statistically significant even after correction for multiple testing. (Appendix Table 17). The results indicate that household size is a strong predictor of electricity consumption, with each additional square meter associated with an approximate 0.4% increase in the annual consumption within the same EPC-label and same number of occupants. Each additional occupant is responsible for around 16% increase in energy consumption fro same EPC class and house size, and this is also supported by larger sum of squares (220.87).(Appendix Table 16) It seems that annual electricity consumption is more behavior-dependent rather than driven by house size. For each extra bedroom with size of 15 m^2, the power consumption increases by 6%. However, each additional household occupant increases electricity consumption by 16%. Since there is no evidence that larger households tend to live in larger dwellings, a plausible explanation for the observed linear relationship between energy consumption and the number of occupants is that energy demand is primarily driven by individual behaviors.

Table 2: Regression results for the final model

	Beta	Std. Error	p-value	OR	CI 2.5%	CI 97.5%
(Intercept)	7.220	0.020	<0.001	1366.238	1313.456	1421.142
sq_meters	0.004	0.000	<0.001	1.004	1.004	1.004
epc_labelB	0.087	0.009	<0.001	1.091	1.071	1.111
epc_labelC	0.157	0.014	<0.001	1.170	1.137	1.203
epc_labelD	0.238	0.015	<0.001	1.268	1.232	1.306
epc_labelE	0.316	0.017	<0.001	1.371	1.325	1.419
epc_labelF	0.394	0.018	<0.001	1.482	1.431	1.535
Built 1970–2000	-0.009	0.009	0.361	0.991	0.973	1.010
Built post-2000	-0.010	0.015	0.531	0.991	0.961	1.020
occupancy_count	0.151	0.001	<0.001	1.163	1.160	1.166

4.2 The income effect

The total annual household income was removed from the model due to not adding any improvement in the model. The income variable kept the adjusted R2 as in the main model, but increased the BIC value by 7 units; suggesting less supported model. (Appendix Table 19) Income appears to be a meaningful predictor of annual power consumption based on real-life considerations and prior knowledge (i.e., higher income is associated with more devices and higher power use). This was significant in the model before adjusting for the floor area. However, after including dwelling size in the model the effect of income attenuated and became non-significant. This could be due to that both covariates are sharing similar information. There is a positive correlation between annual power consumption and total income, as well as between annual power consumption and floor area. This suggests that dwelling size may act as a mediator in the relationship between income and energy consumption, capturing much of the variability that would otherwise be attributed to income. To further investigate this relationship, a conditional analysis was performed by dividing floor area into four equal categories and calculating the correlation between annual consumption

and total income within each group. This analysis showed that, within each floor area group (i.e., conditional on floor area), the correlation between income and annual consumption is very weak. In conclusion, total income has an effect through the floor area. (Figure 12, and 13)

4.3 The impact of a high-quality EPC rating ('A' or 'B') as household occupancy increases.

A model with an interaction term between occupancy count and a dummy variable for high-quality EPC label A-B was fitted to investigate whether the effect of EPC label is dependent on the number of occupants. The correlation between annual electricity consumption and occupancy count was very similar in both EPC label groups. This was also confirmed by that the interaction term was not statistically significant with a p-value of 0.77. This confirm our basic findings, the effect of number of occupants on the annual consumption is 16% for each additional person keeping other predictors fixed. This signifies that the effect of a proper insulation on electricity consumption does not change with the number of occupants.(Table 21)

5 Conclusions and Ethical considerations

The present analysis shows that the EPC label and the number of occupants are the most relevant predictors of annual electricity consumption. Compared to dwelling in energy class A, lower efficiency classes exhibit progressively higher electricity consumption, from about 7% for class B to nearly 48% for class F. Each additional occupant is responsible for a linear increase in electricity consumption of about 16%.

We should keep in mind the model explains 88% of the variance in the dependent variable, adjusted for the number of predictors. The results should be interpreted with caution as the study considers energy consumption in Belgium with no enough information about the date/settings of data collection. The variation in using energy systems between houses was not considered in this analysis, as some people prefer gas and electricity, while others are dependent on electricity only. Furthermore, solar system use may be an unmeasured confounder. Socioeconomic status may affect energy consumption by using different and less effective home systems to overcome some financial obstacles. These factors are not included in this model and should be taken into account while interpreting the results. From the policy-maker perspective, this study provides a model to predict the energy consumption and will not provide a decision to subsidize structural renovations or not, but may support behavioral interventions. It remains to be established whether encouraging the use of common areas (for example, co-working spaces in the building) will have a substantial impact on total energy demand. The heating and cooling system is a "shared" consumption in a household. This may reduce the contribution of the number of occupants in this model. Finally, it can be included that the energy demand is primarily driven by individual behaviors.

6 Appendix

Table 3: Data Dictionary

Name	Description	Type_Coding
annual_kwh	yearly electricity use	Continuous (kWh/year)
sq_meters	indoor living area	Continuous (m^2)
epc_label	energy performance certificate	Ordinal: A, B, ..., F
building_era	construction period	Ordinal: ref = Pre-1970, 1970-2000, post-2000
income_euro	net household income	Continuous (€/year)
occupancy_count	permanent residents	Count: 1, 2, ..., n
dist_to_brussels	distance to Brussels center	Continuous (km)
roof_color	roof tile color	Categorical: red, black, other
pet_ownership	owner has a pet	Binary: 0 = No, 1 = Yes

Table 4: Data Ordered

	annual_kwh	sq_meters	epc_label	building_era	income_euro	occupancy_count
267	5372190.00	156.15	C	1970-2000	44737.59	4
1330	4635780.00	114.28	D	Pre-1970	12830.40	4
1570	4428940.00	154.49	F	Pre-1970	64433.95	1
369	4004330.00	133.92	D	1970-2000	58718.83	2
12	2955140.00	100.34	A	Post-2000	31930.12	3
77	17197.12	268.11	F	Pre-1970	115384.00	7
1038	16395.02	304.18	E	Pre-1970	131393.77	7
1051	14024.23	204.98	F	Pre-1970	81213.77	7
452	13872.60	177.94	F	Pre-1970	73933.89	7
2057	13821.56	162.59	F	Pre-1970	36281.03	7

6.1 Summary tables for missing values

Table 5: Missing data summary

Variable	Missing_n	Missing_pct
annual_kwh	5	0.2
sq_meters	5	0.2
epc_label	5	0.2
income_euro	5	0.2
occupancy_count	5	0.2
dist_to_brussels	5	0.2
building_era	0	0.0
roof_color	0	0.0
pet_ownership	0	0.0

6.2 Summary tables for categorical varibales

Tables below refer to the dataset with only rows 66, 36 135 removed.

Table 6: EPC Label Distribution

epc_label	n	percent
A	554	22.2
B	436	17.5
C	532	21.3
D	391	15.7
E	338	13.5
F	241	9.7
NA	5	0.2

Table 7: Building Era Distribution

building_era	n	percent
1970-2000	856	34.3
Post-2000	814	32.6
Pre-1970	827	33.1

Table 8: Roof Color Distribution

roof_color	n	percent
Black	832	33.3
Other	788	31.6
Red	877	35.1

Table 9: Pet Ownership Distribution

pet_ownership	n	percent
0	1509	60.4
1	988	39.6

Table 10: Occupancy Count Distribution

occupancy_count	n	percent
1	352	14.1
2	357	14.3
3	346	13.9

4	379	15.2
5	377	15.1
6	342	13.7
7	339	13.6
NA	5	0.2

6.3 Summary tables for continuous varibales

Tables below refer to the dataset with only rows 66, 36 135 removed and the dataset after NA and outliers removed.

Table 11: Summary Statistics for Continuous Variables (row data)

Statistic	annual_kwh	sq_meters	dist_to_brussels	income_euro
Mean	14445.58	158.19	63.03	48844.80
SD	194579.20	36.47	33.07	22504.32
Median	5432.60	155.84	63.24	44557.61
IQR	2996.35	46.80	57.70	25871.05
Min	1824.63	60.26	5.11	8487.91
Max	5372190.00	341.30	120.00	206367.91

Table 12: Summary Statistics for Continuous Variables (cleaned dataset)

Statistic	annual_kwh	sq_meters	dist_to_brussels	income_euro
Mean	14445.58	158.19	63.03	48844.80
SD	194579.20	36.47	33.07	22504.32
Median	5432.60	155.84	63.24	44557.61
IQR	2996.35	46.80	57.70	25871.05
Min	1824.63	60.26	5.11	8487.91
Max	5372190.00	341.30	120.00	206367.91

6.4 Summary plots for continuous varibales

Figures below refer to the dataset used for the regression model from which missing values and outliers in annual energy consumption were removed as explained in section two. From the box plot we observe that several variables exhibit right-skewness and upper-tail outliers, particularly income and energy consumption. While annual energy consumption deviates significantly from normality, its log transformed analogue follows an approximate normal distribution dispute a mild deviation from the expected quantiles in the lower and upper tail. The QQ plot for floor area shows only mild departures from normality. The average distance to Brussels is 62.96 km. Its QQ plot indicates noticeable curvature, implying that the distribution deviates from normality, possibly reflecting clustering of households at certain distance ranges during the sampling procedure. Household income QQ plot shows strong upper-tail deviation, suggesting positive skewness driven by high-income outliers.

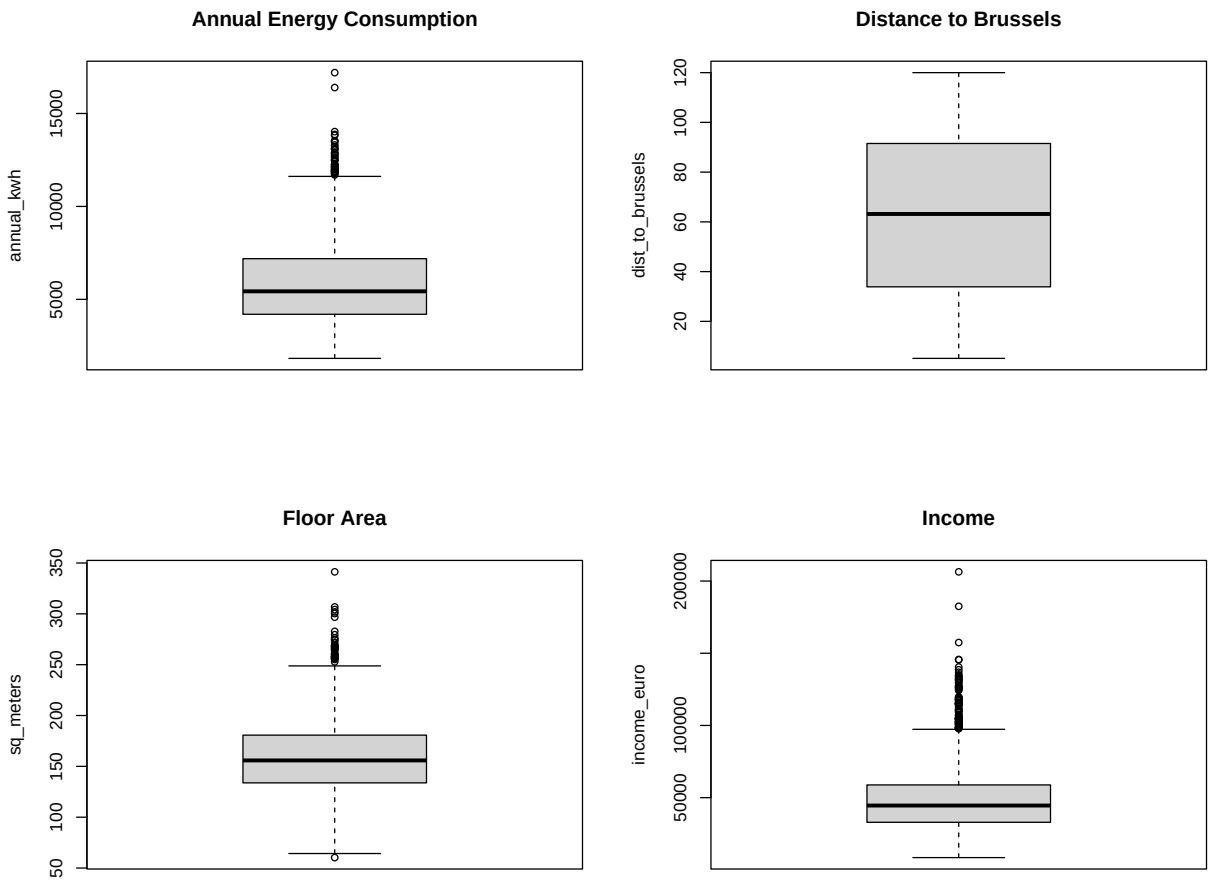


Figure 2: Box plots for continuous variables

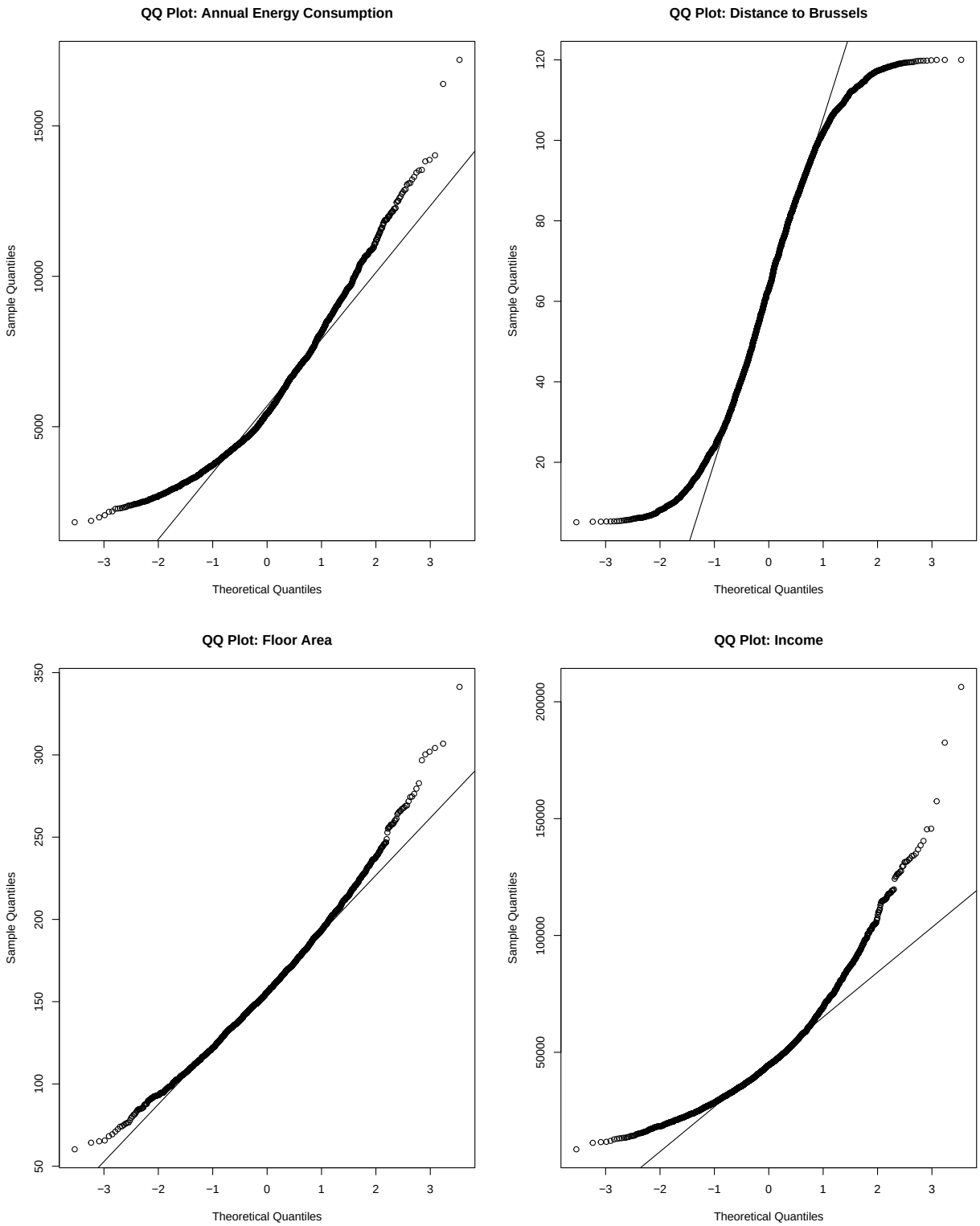


Figure 3: QQ plots for continuous variables

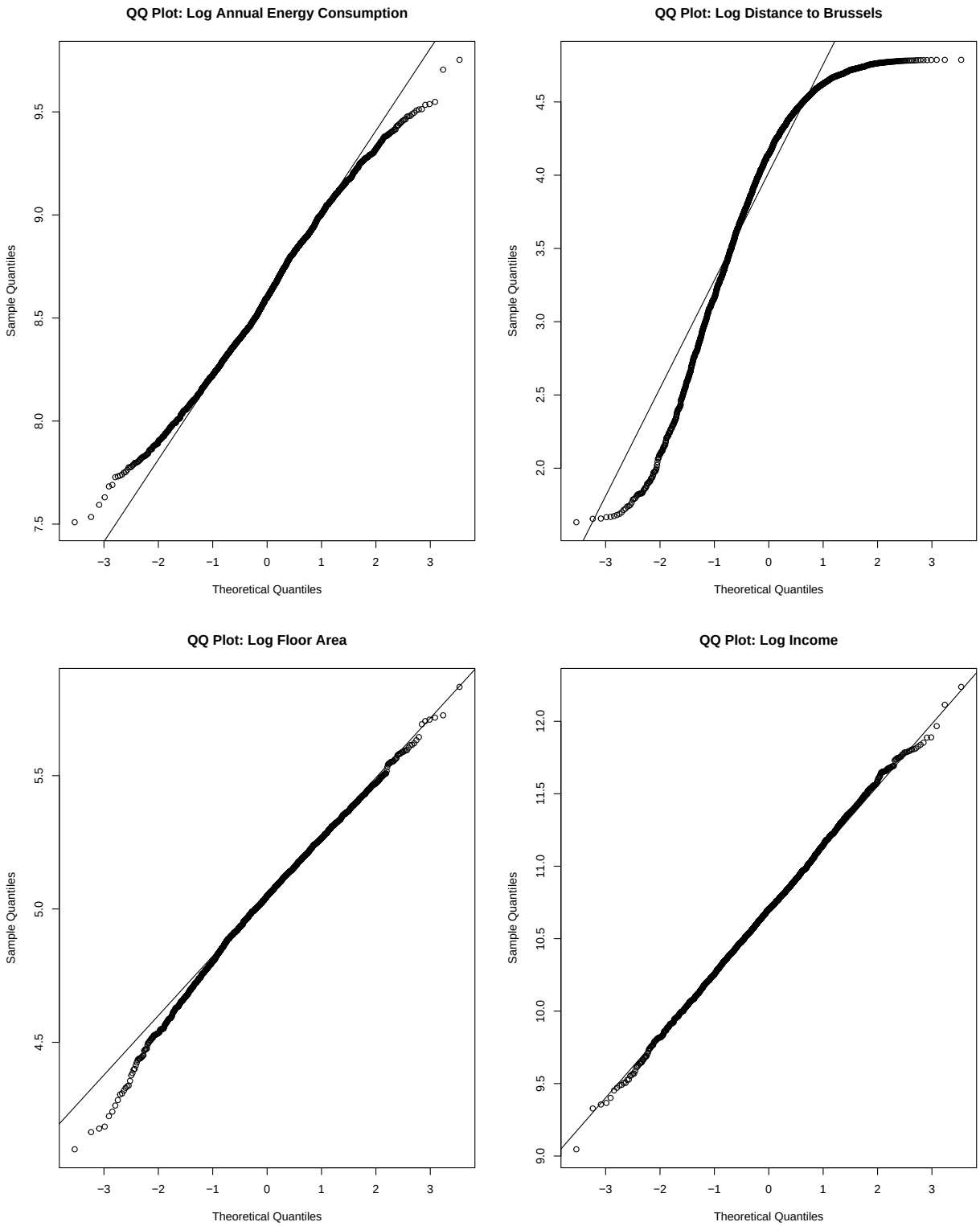
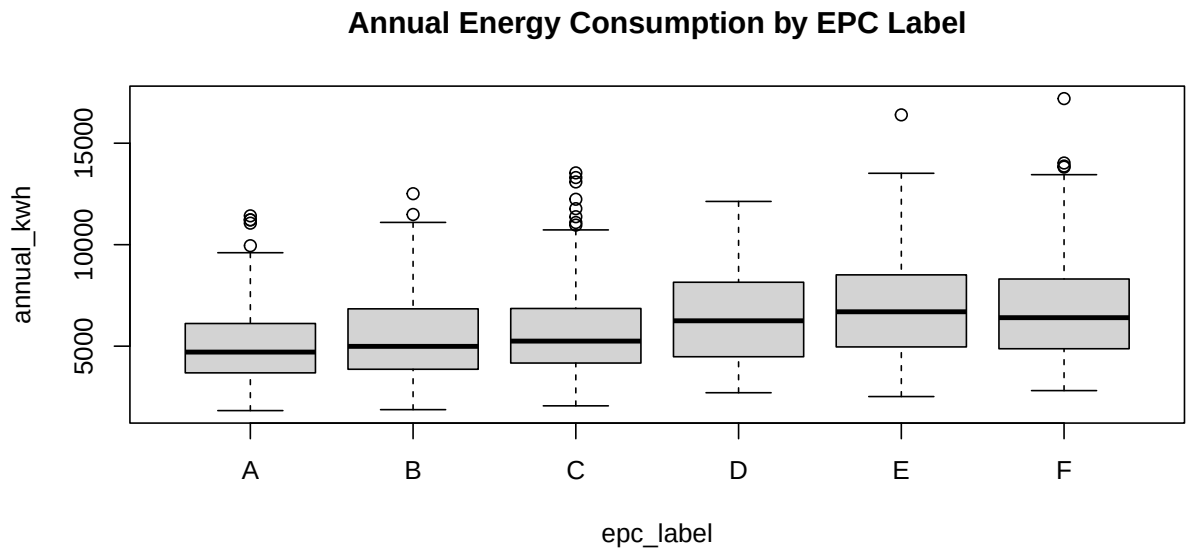


Figure 4: QQ plots for log-transformed continuous variables

6.5 Relationship between Annual Energy Consumption and non-continuous variables.

Among categorical and ordinal variables, EPC label and occupancy count are the most correlated to annual energy consumption. In addition, EPC label and building era are correlated with each other since the most efficient labels are clustered in the two most recent building eras and most recent buildings. Therefore, building era might play as a confounder in the model.



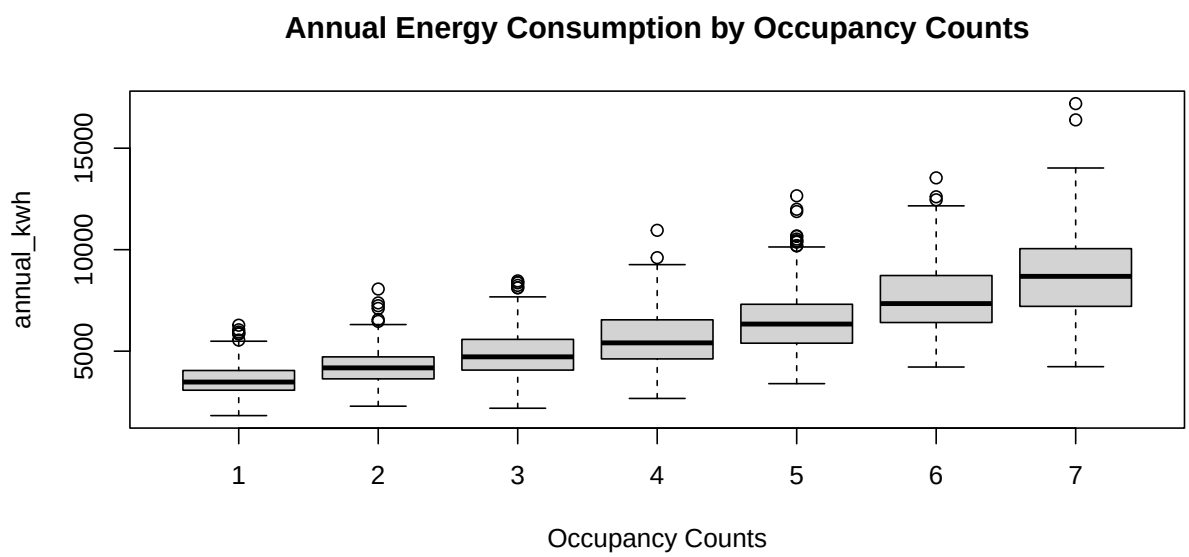
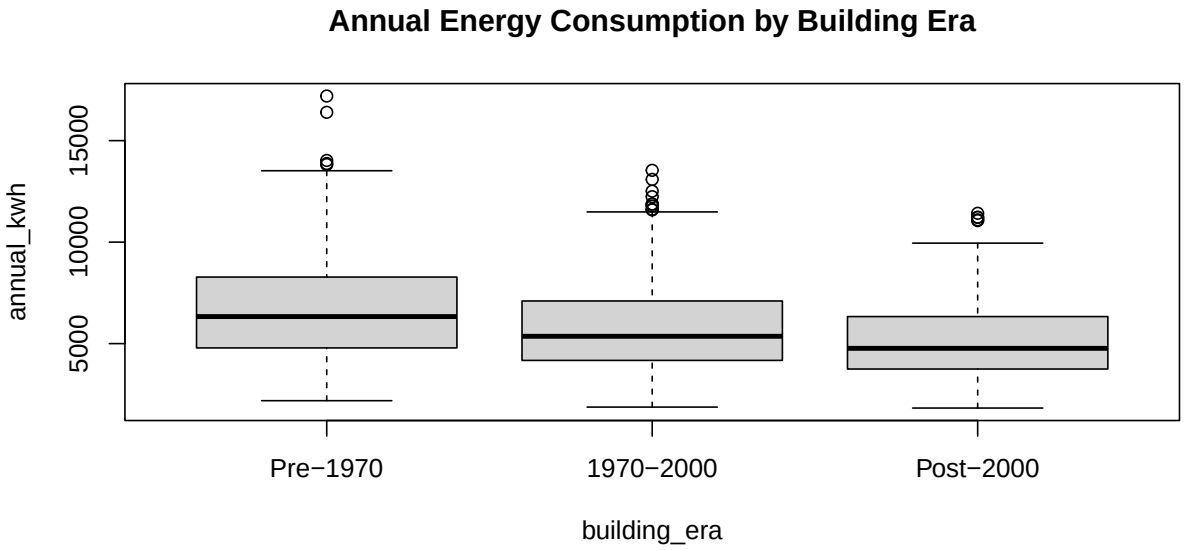


Figure 5: Box plots for annual energy consumption vs categorical variables (part 1)



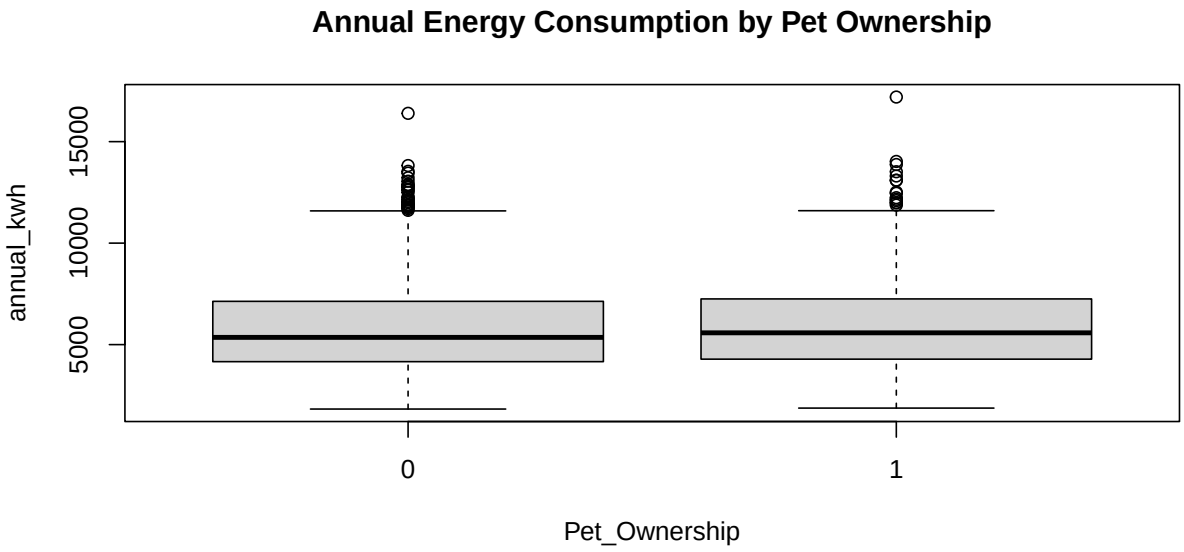


Figure 6: Box plots for annual energy consumption vs categorical variables (part 2)

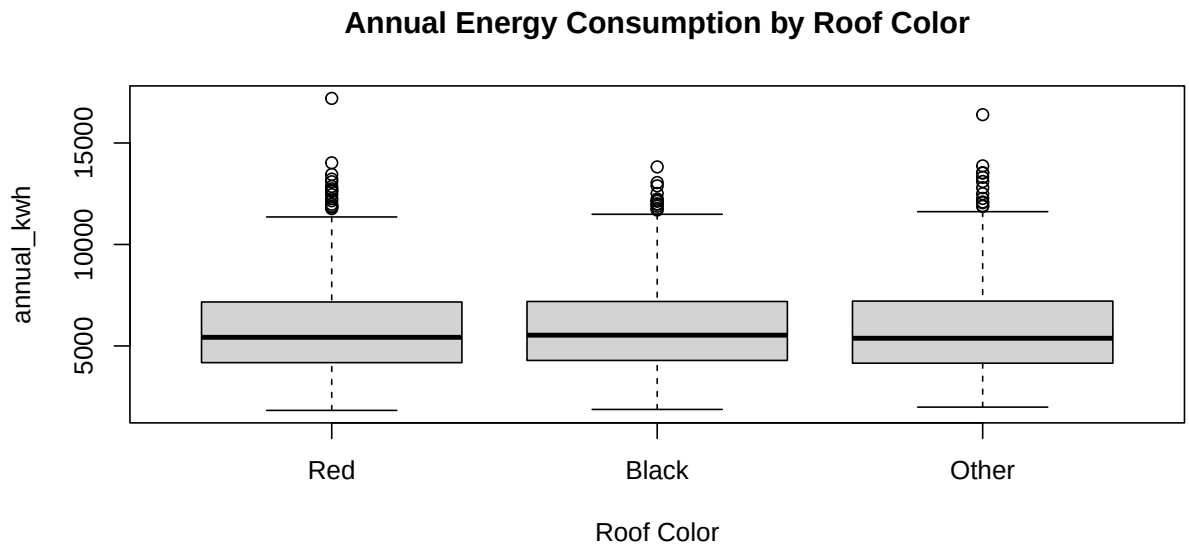
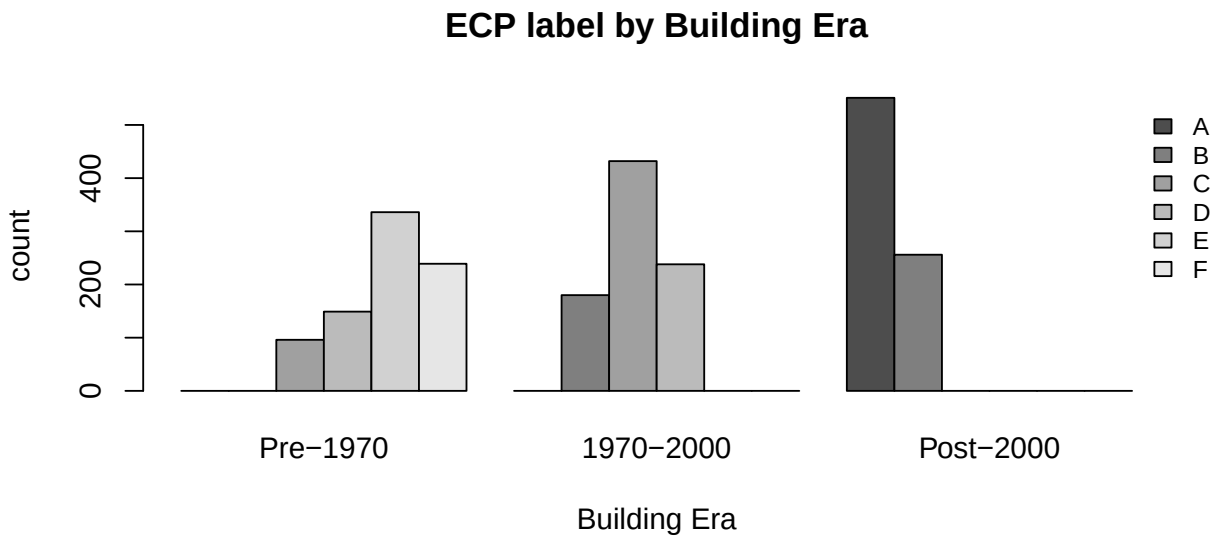


Figure 7: Box plots for annual energy consumption vs categorical variables (part 3)

6.6 Causal Relationships Among Variables

Building era was treated as a confounder in the relationship between EPC label and electricity consumption since it is related to both: the main predictor and the outcome variable.

Income was treated as a mediator in the relationship between EPC electricity consumption and size of the households since people with higher income can afford larger dwellings which in turn consume more electricity. Indeed, the income and apartment size are strongly correlated (rho=0.72).



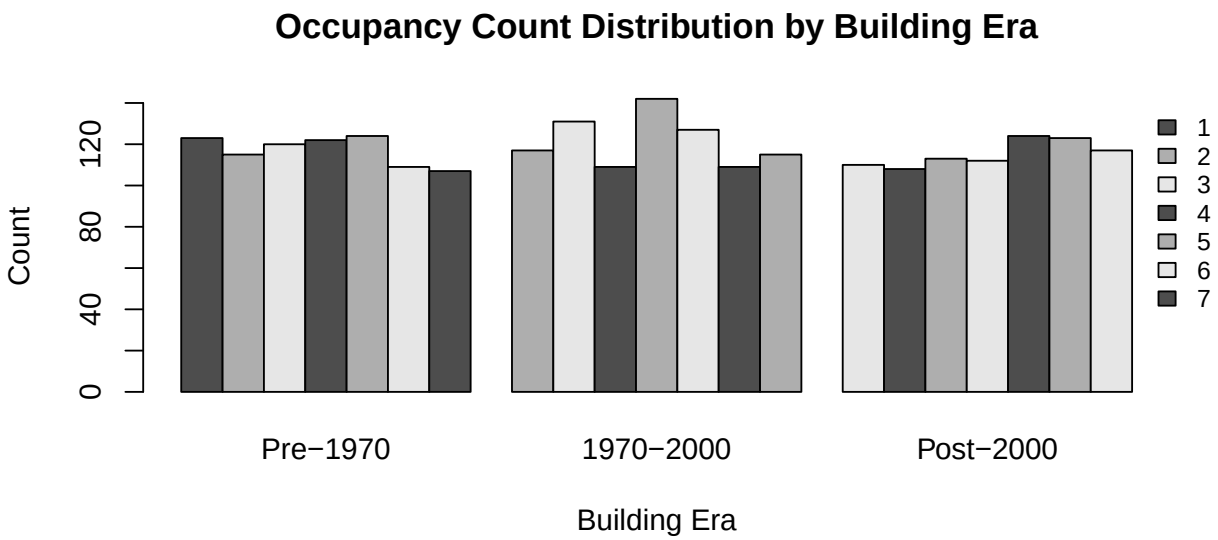


Figure 8: Bar plots predictors against each other (selected cases)

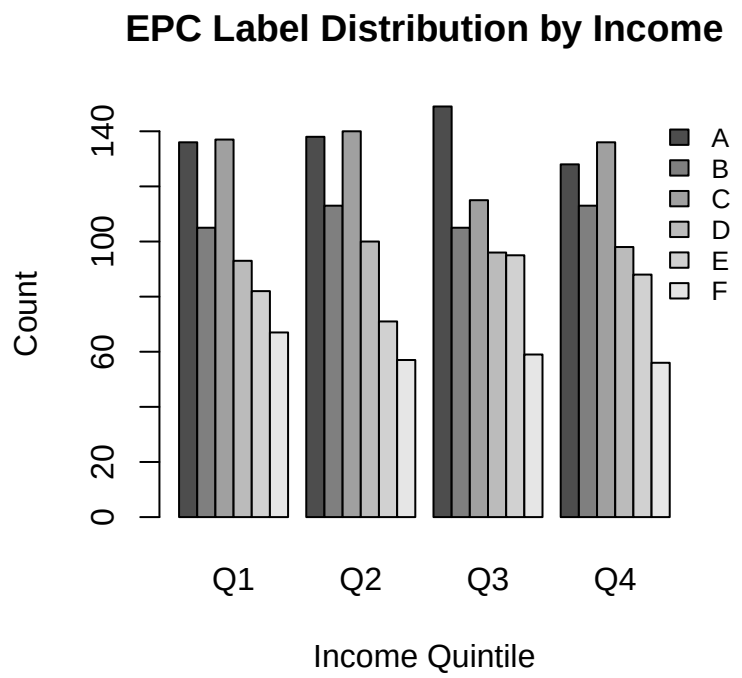


Figure 9: Bar plots of EPC label class by Income percentile

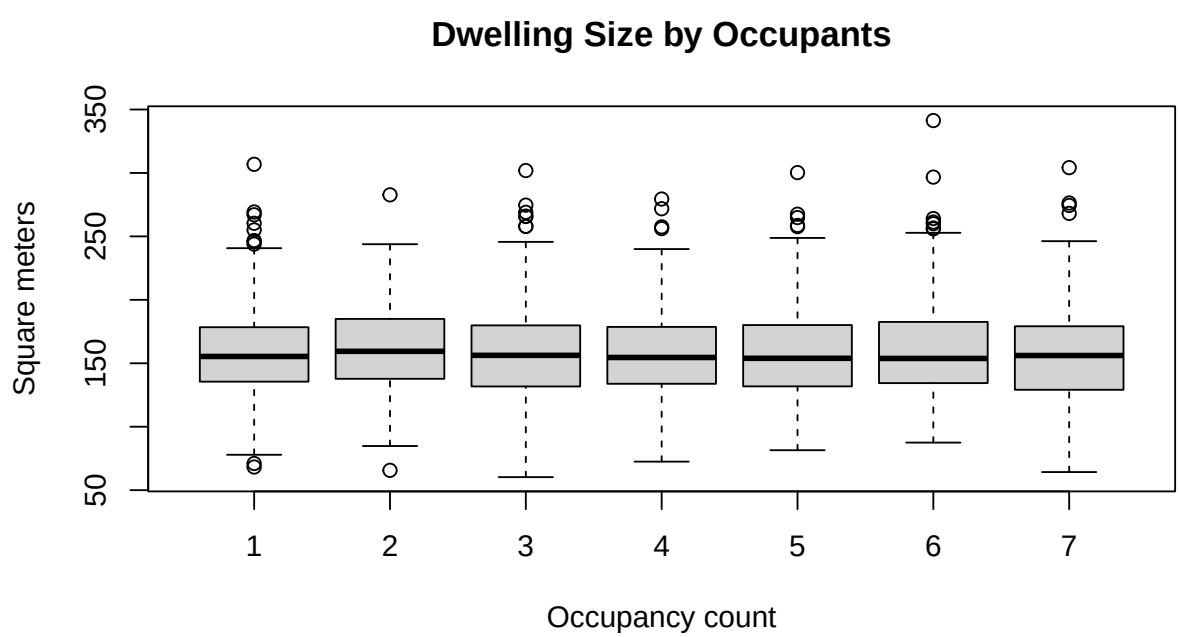


Figure 10: Box plots of dwelings square meters by occupancy count

Table 13: Mean Annual Energy Consumption by Size and Income Quartiles

	Income Quintiles			
	Q1	Q2	Q3	Q4
Q1	4812	5022	5085	5223
Q2	5497	5422	5473	5804
Q3	6190	6068	6023	6236
Q4	6595	6288	6606	7208

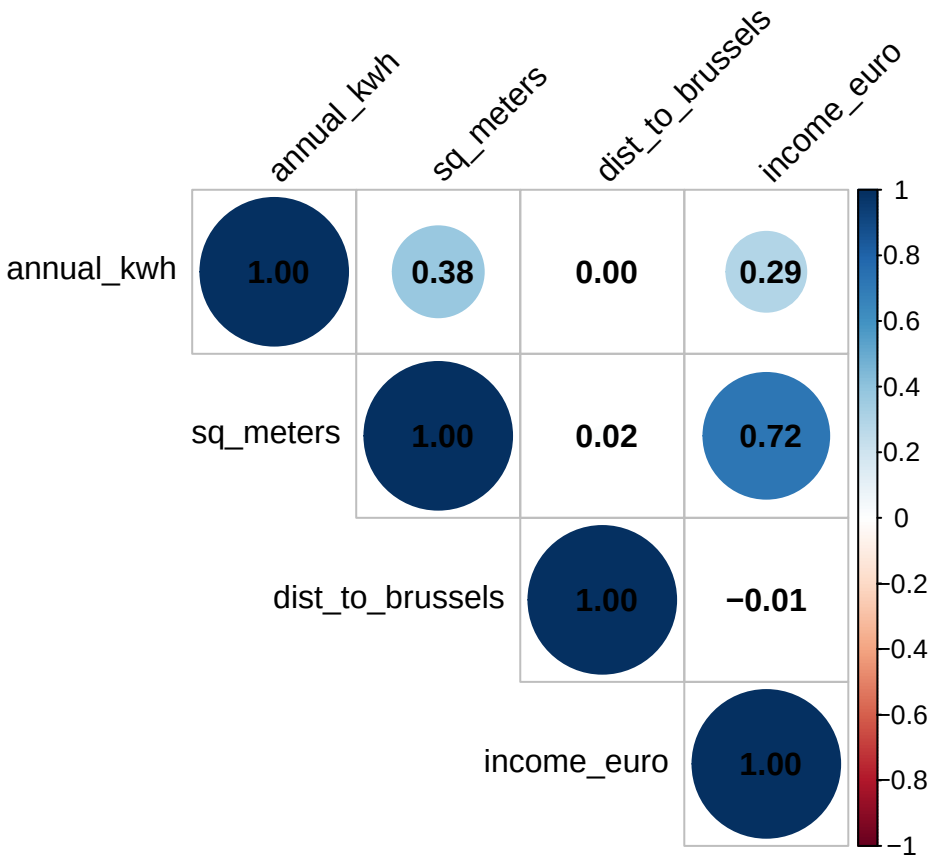


Figure 11: Correlation plots for continuous variables

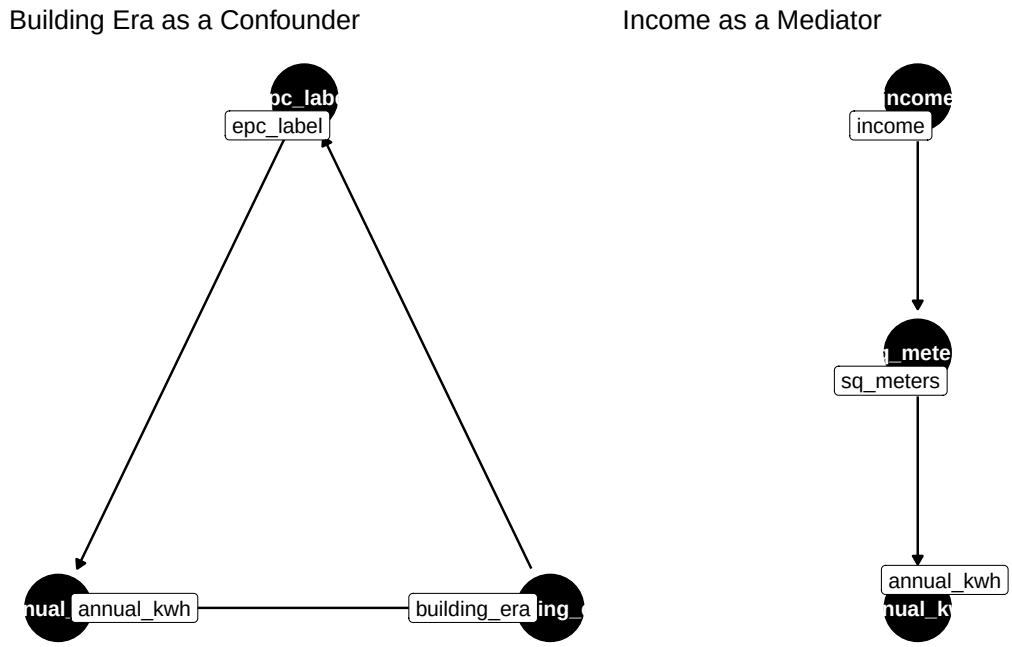


Figure 12: Directed acyclic graphs (DAGs)

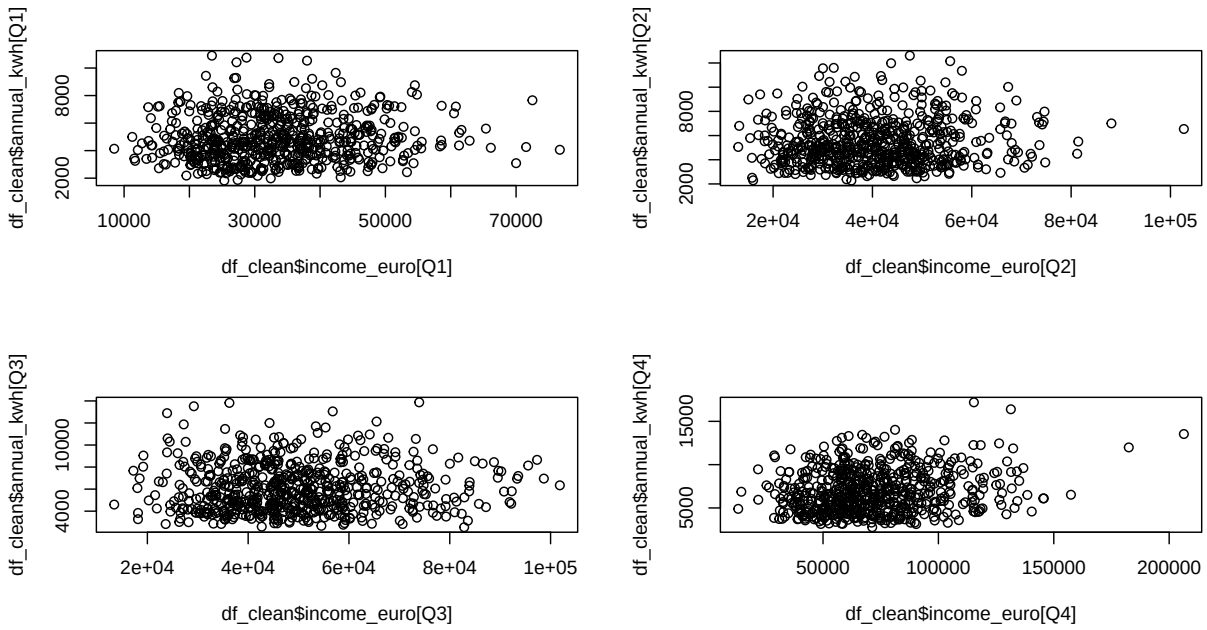


Figure 13: Relationship between Income and Energy Consumption per House size quartiles

6.7 Regression Model

The initial regression model was the following:

$$
\begin{aligned}
\text{annual_kwh}_i = \beta_0 &+ \beta_1\,\text{sq_meters}_i + \beta_2\,\text{income_euro}_i + \beta_3\,\text{occupancy_count}_i + \beta_4\,\text{dist_to_brussels}_i \\
&+ \boldsymbol{\beta}_5\,\text{epc_label}_i + \boldsymbol{\beta}_6\,\text{building_era}_i + \boldsymbol{\beta}_7\,\text{roof_color}_i + \beta_8\,\text{pet_ownership}_i \\
&+ \beta_9(\text{income_euro}_i \times \text{occupancy_count}_i) \\
&+ \boldsymbol{\beta}_{10}(\text{epc_label}_i \times \text{sq_meters}_i) \\
&+ \boldsymbol{\beta}_{11}(\text{epc_label}_i \times \text{occupancy_count}_i) \\
&+ \boldsymbol{\beta}_{12}(\text{epc_label}_i \times \text{income_euro}_i) \\
&+ \boldsymbol{\beta}_{13}(\text{occupancy_count}_i^2) + \varepsilon_i
\end{aligned}
$$

The final model retained the following predictors:

$$
\log \text{annual_kwh}_i = \beta_0 + \beta_1\,\text{sq_meters}_i + \beta_2\,\text{occupancy_count}_i + \beta_3\,\text{epc_label}_i + \beta_4\,\text{building_era}_i + \varepsilon_i
$$

6.7.1 Regression Results Summary Tables: Initial vs Final model

Table 14: Regression results for the full model (starting model)

	Estimate	Std. Error	t value	p-value
(Intercept)	7.21014305395381	0.0130388929287292	552.972027101116	<0.001
sq_meters	0.00395819092169818	6.77098595738489e-05	58.458117423521	<0.001
epc_labelB	0.087431994802567	0.00787211999221748	11.1065373608385	<0.001
epc_labelC	0.159161541433674	0.00748346002244085	21.2684428000406	<0.001
epc_labelD	0.241755849618861	0.00814490329903896	29.6818563392135	<0.001
epc_labelE	0.325440608379095	0.00850069156573529	38.2840155842009	<0.001
epc_labelF	0.403114597811007	0.00953189792886413	42.2911156644168	<0.001
occupancy_count	0.151099953320912	0.00125053256132219	120.82848379506	<0.001

Table 15: Regression results for the final model (with 95% CI)

	Estimate	Std. Error	t value	p-value	CI 2.5%	CI 97.5%
Intercept	7.220	0.020	359.335	<0.001	7.180	7.259
sq_meters	0.004	0.000	58.429	<0.001	0.004	0.004
epc_labelB	0.087	0.009	9.369	<0.001	0.069	0.105
epc_labelC	0.157	0.014	11.031	<0.001	0.129	0.185
epc_labelD	0.238	0.015	15.933	<0.001	0.208	0.267
epc_labelE	0.316	0.017	18.129	<0.001	0.282	0.350
epc_labelF	0.394	0.018	21.922	<0.001	0.358	0.429
building_era1970-2000	-0.009	0.009	-0.913	0.361	-0.027	0.010

	Estimate	Std. Error	t value	p-value	CI 2.5%	CI 97.5%
building_eraPost-2000	-0.010	0.015	-0.627	0.531	-0.039	0.020
occupancy_count	0.151	0.001	120.796	<0.001	0.149	0.154

Table 16: Anova Table

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
sq_meters	1	48.59	48.59	3221.69	0.0
epc_label	5	36.03	7.21	477.69	0.0
building_era	2	0.03	0.01	0.91	0.4
occupancy_count	1	220.09	220.09	14591.69	0.0
Residuals	2467	37.21	0.02	NA	NA

Table 17: Contrast results for the EPC Labels

	OR	CI_2.5	CI_97.5	p.value
B - A	1.0909	1.0667	1.1158	<0.001
C - A	1.1696	1.1300	1.2106	<0.001
D - A	1.2681	1.2231	1.3147	<0.001
E - A	1.3715	1.3148	1.4307	<0.001
F - A	1.4823	1.4191	1.5482	<0.001

Table 18: VIF result after including the income

	GVIF	Df	GVIF^(1/(2*Df))
sq_meters	2.117061	1	1.455012
epc_label	12.820059	5	1.290592
building_era	12.701413	2	1.887830
occupancy_count	1.020948	6	1.001729
income_euro	2.117228	1	1.455070

Table 19: Model godness after inclusing income as a regressor

	df	BIC
modelincom	17	-3238.844
final_model	11	-3283.627

Table 20: Regression model including income

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	7.373	0.021	358.961	0.0000000
sq_meters	0.004	0.000	39.763	0.0000000
epc_labelB	0.087	0.009	9.341	0.0000000
epc_labelC	0.157	0.014	11.016	0.0000000
epc_labelD	0.238	0.015	15.917	0.0000000
epc_labelE	0.317	0.017	18.129	0.0000000
epc_labelF	0.394	0.018	21.913	0.0000000
building_era1970-2000	-0.008	0.009	-0.866	0.3864698
building_eraPost-2000	-0.009	0.015	-0.584	0.5593577
occupancy_count2	0.146	0.009	15.745	0.0000000
occupancy_count3	0.301	0.009	32.213	0.0000000
occupancy_count4	0.459	0.009	50.227	0.0000000
occupancy_count5	0.602	0.009	65.784	0.0000000
occupancy_count6	0.752	0.009	80.226	0.0000000
occupancy_count7	0.906	0.009	96.611	0.0000000
income_euro	0.000	0.000	0.609	0.5424017

Table 21: Results Interaction between EPC Label and Occupancy Count

	Estimate	Std. Error	t value	p-value
(Intercept)	7.6770	0.0151	510.0345	<0.001
occupancy_count2	0.1580	0.0129	12.2821	<0.001
occupancy_count3	0.3038	0.0130	23.3722	<0.001
occupancy_count4	0.4661	0.0127	36.7655	<0.001
occupancy_count5	0.5914	0.0127	46.5063	<0.001
occupancy_count6	0.7620	0.0133	57.4203	<0.001
occupancy_count7	0.9087	0.0132	69.0967	<0.001
epc_high	-0.0913	0.0178	-5.1266	<0.001
sq_meters	0.0039	0.0001	53.2163	<0.001
building_era1970-2000	-0.1290	0.0070	-18.4861	<0.001
building_eraPost-2000	-0.1895	0.0131	-14.4834	<0.001
occupancy_count2:epc_high	-0.0149	0.0208	-0.7183	0.473
occupancy_count3:epc_high	-0.0027	0.0209	-0.1285	0.898
occupancy_count4:epc_high	-0.0152	0.0205	-0.7443	0.457
occupancy_count5:epc_high	0.0149	0.0204	0.7297	0.466
occupancy_count6:epc_high	-0.0196	0.0208	-0.9436	0.345
occupancy_count7:epc_high	-0.0108	0.0209	-0.5180	0.604

7 References

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